

Unit 3 Combinational Circuit Analysis and Design Procedure.

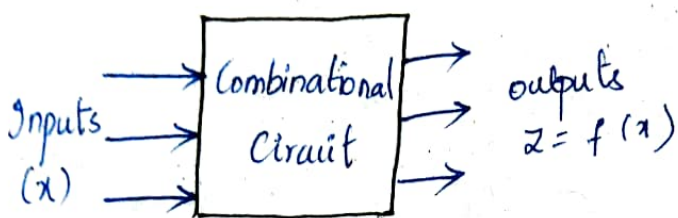


fig. Combinational circuit.

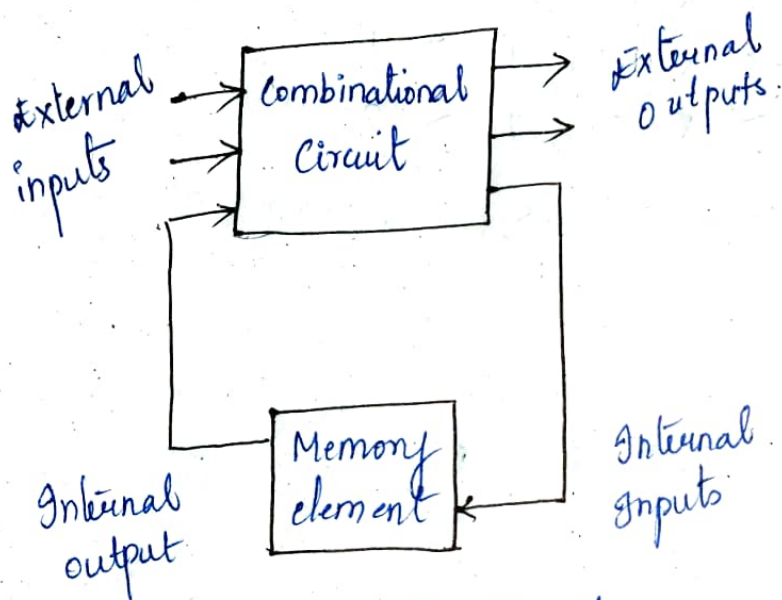


fig. Sequential Circuit

⇒ A sequential circuit are built using combination circuits and memory elements called flip flops. These circuits generate op that depends on the current and the previous state.

Design Procedure of Combinational circuit.

Step 1: Determine the require inputs and outputs and assigned a symbol to each.

Step 2:- derive the truth table that defines the required relationship b/w i/p & o/p.

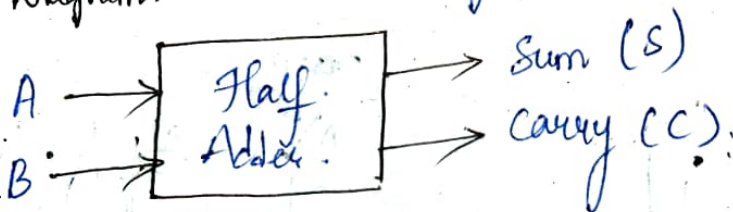
Step 3:- obtain the simplified boolean func. for each o/p as a function of the i/p variables.

Step 4:- draw the logic diagram and verify the correctness of the design.

Half Adder:- (2 inputs, 2 outputs)

no carry.

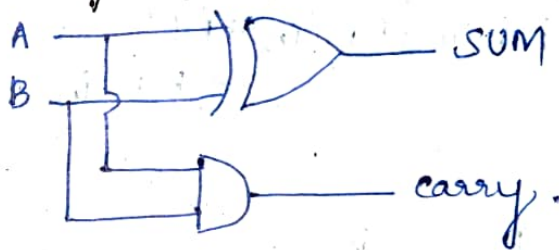
Block Diagram:



Truth Table.

Inputs		Outputs	
A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Logic Diagram.

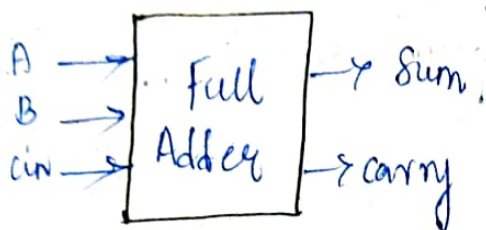


Boolean Expression. $A \oplus B \Rightarrow \text{sum}$
 $A \cdot B \Rightarrow \text{carry}$

The main limitation of half adder are its inability to handle a carry in bit which prevents it from performing multibit addition. It is only capable of adding 2 single bits at a time binary.

Full Adder :-

Block Diagram



Truth table

I/P			o/p	
A	B	Cin	sum	carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Sum

B\Cin	00	01	11	10
A				
0		1		1
1	1		1	

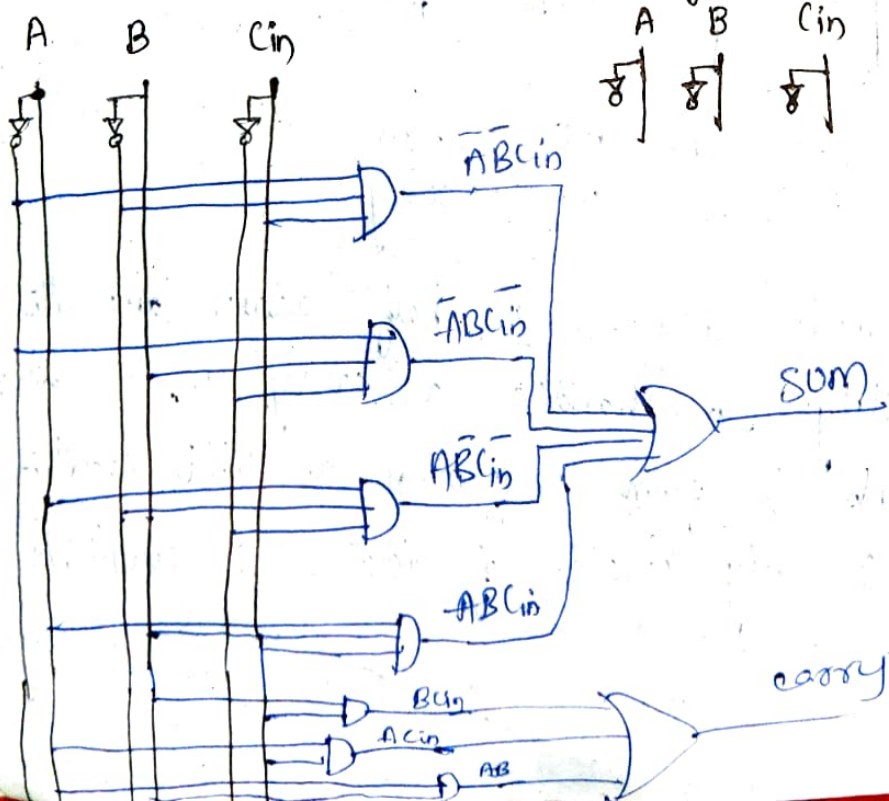
$$\text{sum} = \bar{A}\bar{B}C_{in} + \bar{A}B\bar{C}_{in} + A\bar{B}\bar{C}_{in} + ABC_{in}$$

carry

B\Cin	00	01	11	10
A				
0			1	
1		1	1	1

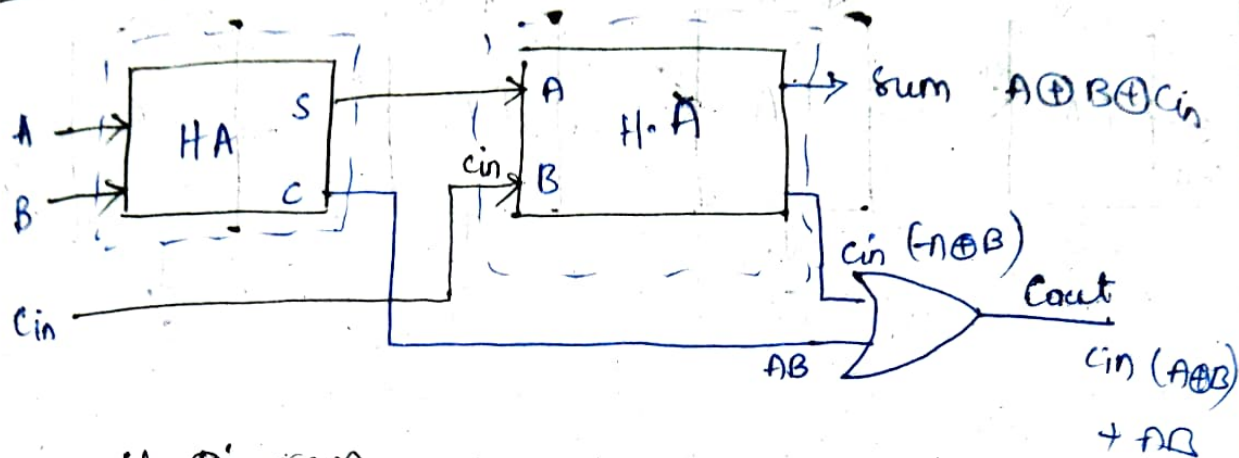
$$B C_{in} + A C_{in} + AB$$

⇒ AOI logic (AND OR Invert)

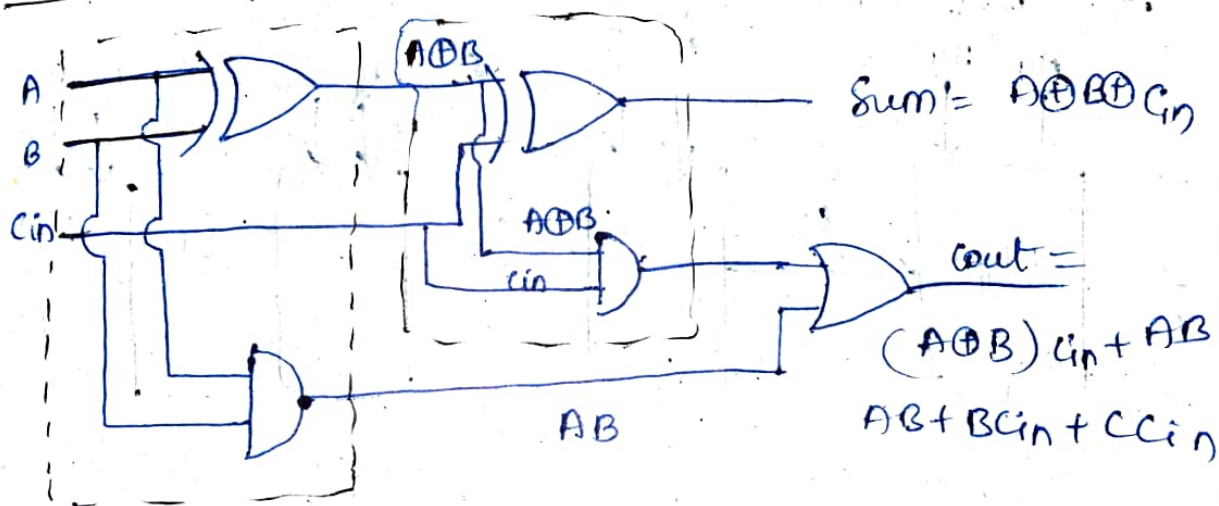


Full Adder using 2 half Adders:-

Block diagram :-



Circuit Diagram



Full Adder = 2 half adders & one OR gate

$$Cout = (A \oplus B) Cin + AB$$

$$= (\bar{A}B + A\bar{B})Cin + AB = \bar{A}BCin + A\bar{B}Cin + AB$$

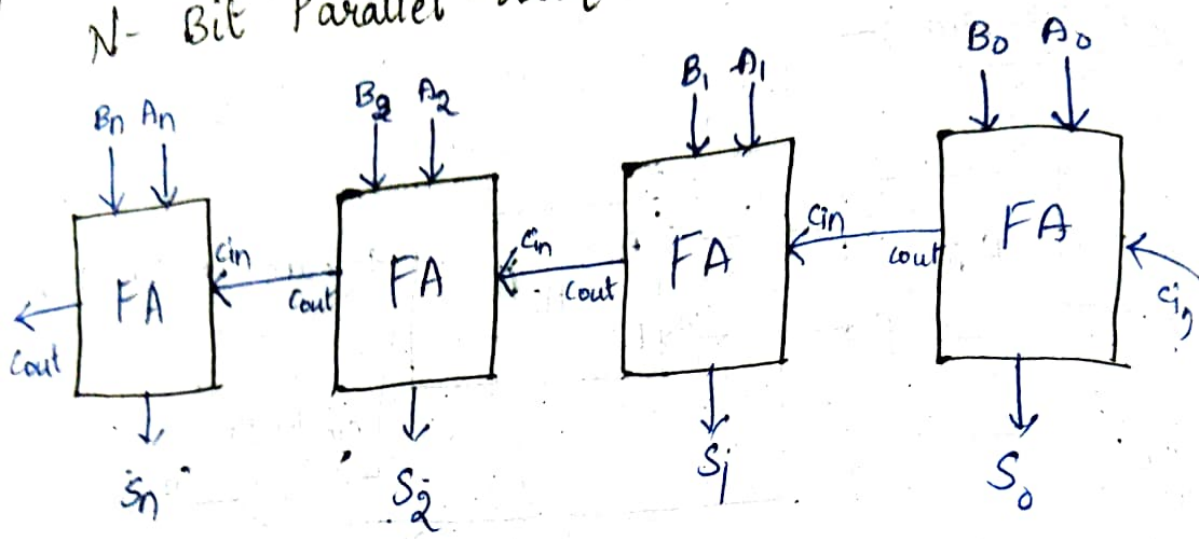
$$= \bar{A}BCin + A\bar{B}Cin + AB = \bar{A}BCin + A(\bar{B}Cin + B)$$

$$= \bar{A}BCin + A(B + Cin) = \bar{A}BCin + AB + ACin$$

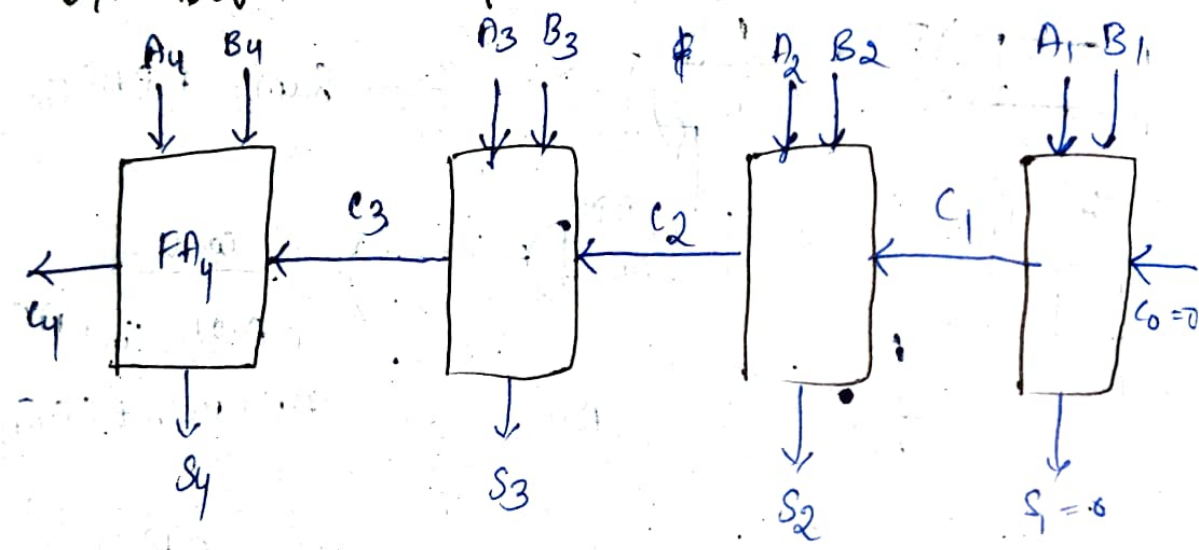
$$= B(\bar{A}Cin + A) + ACin = B(A + Cin) + ACin$$

$$= AB + BCin + ACin$$

N- Bit Parallel Adder.

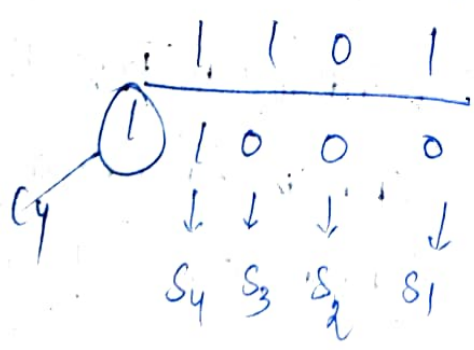


4- Bit Parallel Adder.

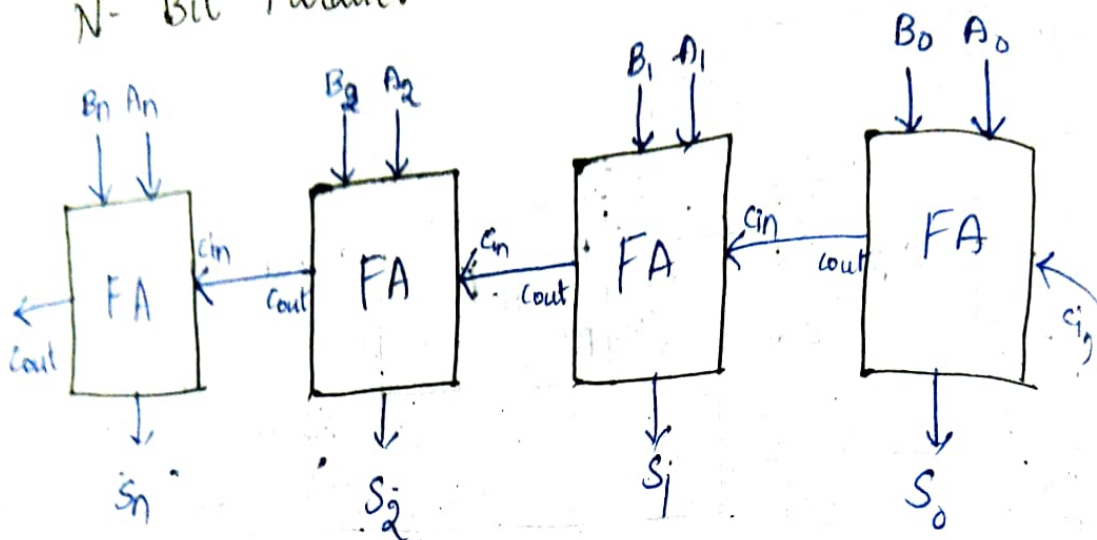


Output:- $c_4 S_4 S_3 S_2 S_1$

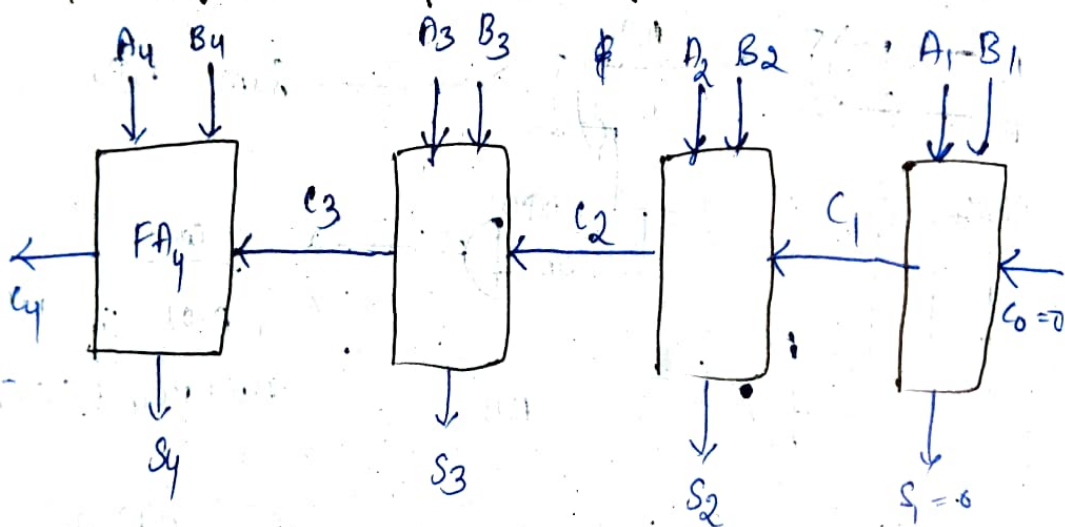
Ex:- 1 0 1 1



N- Bit Parallel Adder.

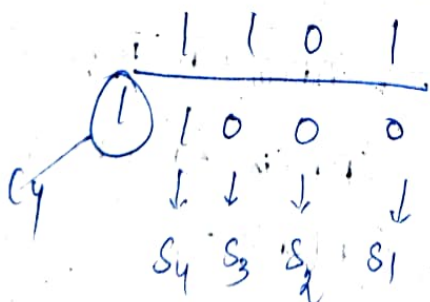


4- Bit Parallel Adder.

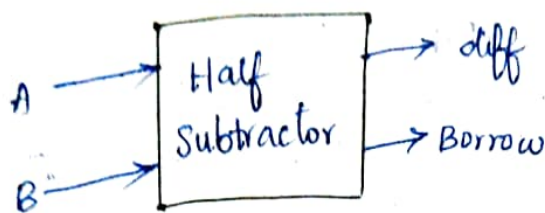


Output:- $C_4 S_4 S_3 S_2 S_1$

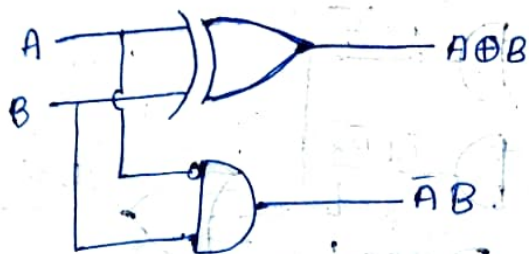
Ex:- 1 0 1 1



Half Subtractor



Logic Diagram



A	B		
	0	1	
0	0	1	Diff
1	1	0	

$\Rightarrow \bar{A}B + A\bar{B}$

A	B	diff	Borrow
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

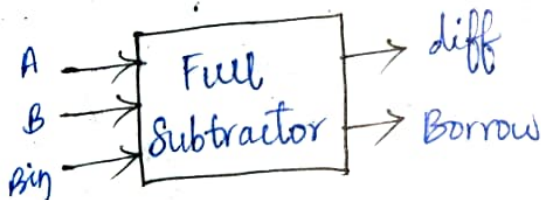
$$\text{diff} \Rightarrow \bar{A}B + A\bar{B} = A \oplus B$$

$$\text{Borrow} \Rightarrow \bar{A}B$$

A	B		
	0	1	
0	0	1	Borrow
1	0	0	

$\Rightarrow \bar{A}B$

Full Subtractor



A	BBin			
	00	01	11	10
0	0	1	1	1
1	1	0	0	0

$$\text{Diff} \Rightarrow \bar{A}\bar{B}B_{in} + \bar{A}B\bar{B}_{in} + A\bar{B}\bar{B}_{in} + AB\bar{B}_{in}$$

A	BBin			
	00	01	11	10
0	0	1	1	1
1	0	0	1	0

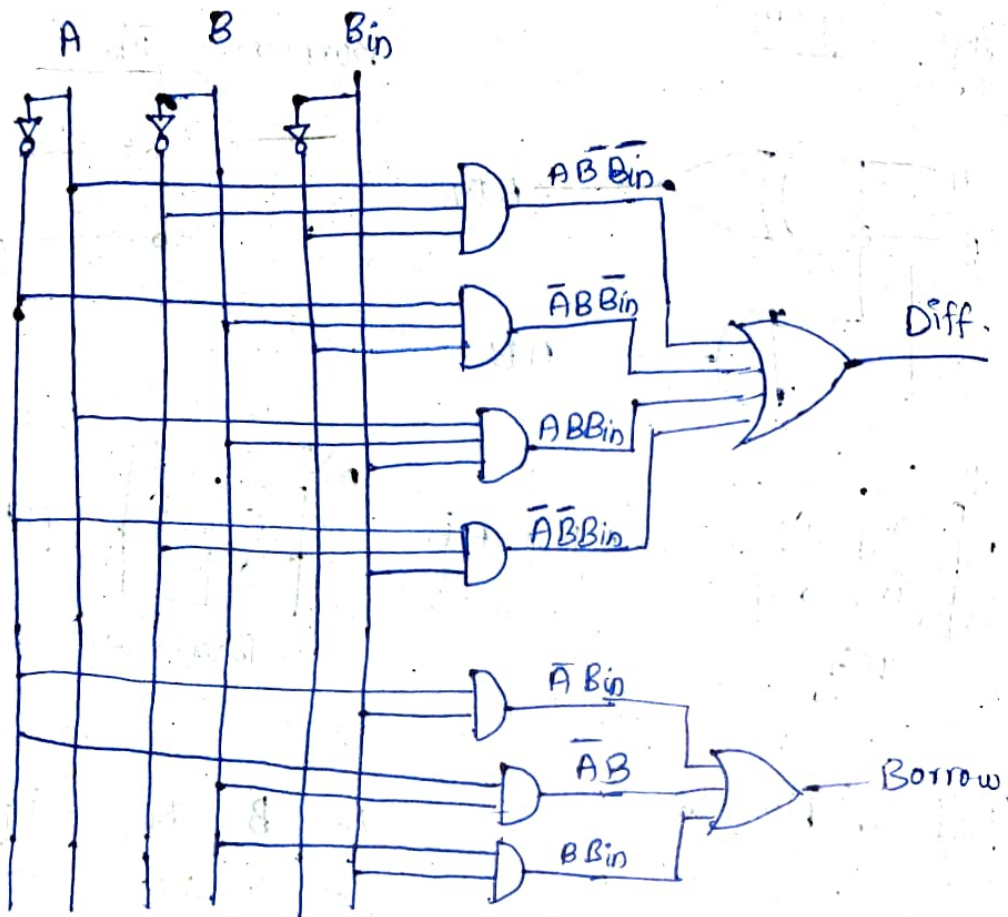
$$\bar{A}B_{in} + \bar{A}B + B\bar{B}_{in}$$

A	B	Bin	Diff	Borrow
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

Diff:- $A\bar{B}\bar{B}_{in} + \bar{A}B\bar{B}_{in} + ABB_{in} + \bar{A}\bar{B}B_{in}$

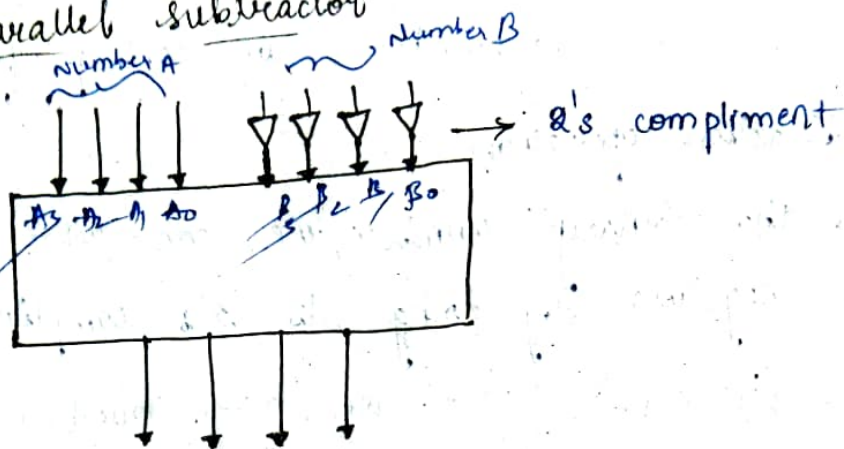
Borrow:- $\bar{A}B_{in} + \bar{A}B + BB_{in}$

Logic Diagram circuit.



N-bit Parallel Subtractor

4-bit Parallel Subtractor



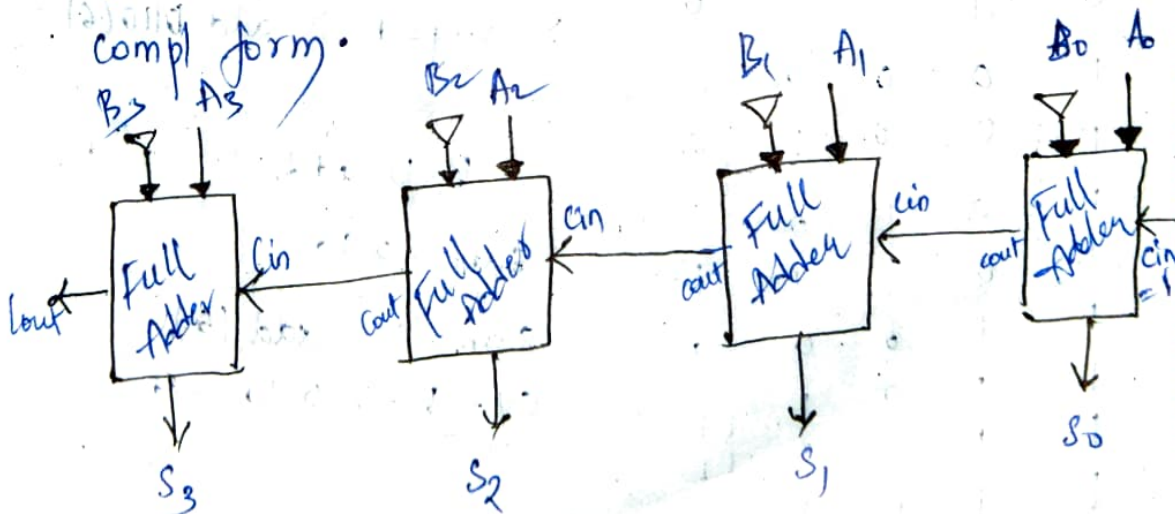
→ The no. to be subtracted (B) is first passed through inverters to obtain its 1's compl.

→ The 4 bit adder then adds A and 2's compl of B to produce subtraction s_3, s_2, s_1, s_0 represent the result of binary subtraction (A-B) and carry output Cout represent the polarity of result.

→ If $A > B$ then $\text{Cout} = 1$ and the result is +ve

→ If $A < B$ then $\text{Cout} = 0$ and result is -ve

Compl form.



- If we want to subtract using adder we can perform the operation $D = A + (-B)$
- If we use -ve numbers, we can perform subtraction operation
- In the binary number, we can represent +ve no to negative by doing its 2's complement operation
- In 2's compl first we make number compl of original no and we add binary 1 in it so

$$-B = \overline{B} + 1$$

BCD Adder / Decimal Adder

i/p				o/p
S_3	S_2	S_1	S_0	Y
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	1
1	0	1	1	1
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	1

BCD Adder

→ If the addition result is 9 or less than 9 then the o/p is correct / valid

→ But if it is > 9
(add 6) add 0110

→ carry = 1 → add 0110 (6)

BCD add

sum 0 to 9

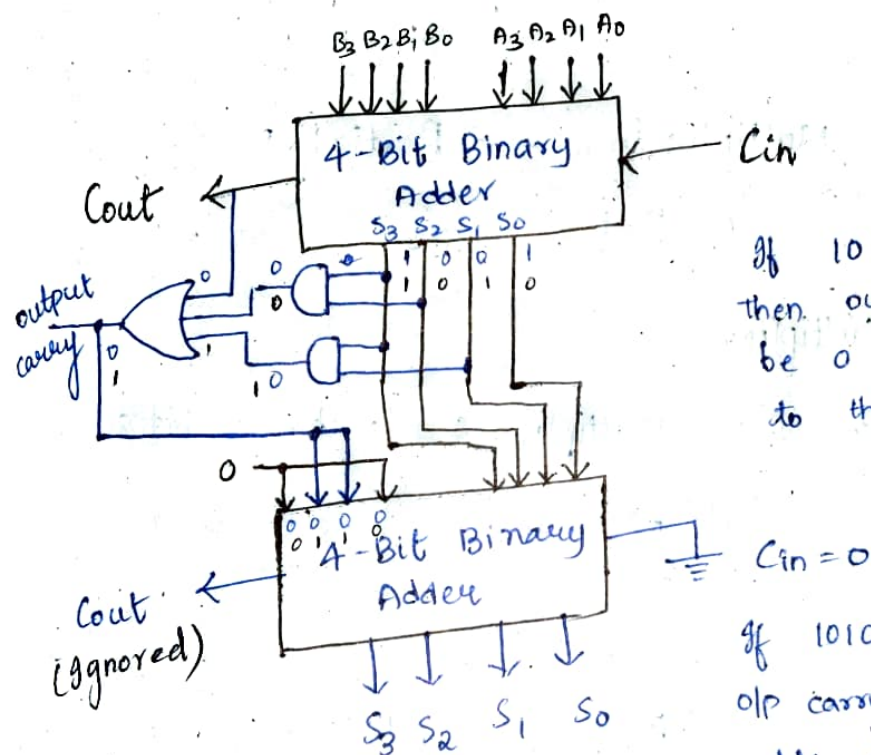
sum > 9 - add (6)

CV = 1 → 0110 add

$S_3 S_2$ \ $S_1 S_0$	00	01	11	10
00	0	0	0	0
01	0	0	0	0
11	1	1	1	1
10	0	0	1	1

$$\cancel{S_0 S_1} + \cancel{S_0 S_2}$$

$$S_3 S_2 + S_3 S_1$$



If 1001 is sum then output carry will be 0 and added 0 to the result

If 1010 is sum then o/p carry = 1 then it adds 6 to result

- The digital systems handle the decimal number in the form of binary coded decimal numbers (BCD)
- A BCD adder circuit that adds two BCD digits & produces a sum digit also in BCD.
- BCD numbers are 10 digits (0 to 9) are represented in binary form: 0000 to 1001

Binary Multiplier:

Digital Binary Multiplier:

- A binary multiplier is a combinational logic circuit used for multiplying a binary no's.
- can be various bitsize $\leftarrow \boxed{\text{multiplicand} \times \text{multiplier}} = \text{product}$
- Bit size of product = sum of size of multiplicand & multiplier.
- step 1: Bitwise Multiplication $\rightarrow$ Partial Product
- step 2: Adding all partial products to get product.

2x2 . BIT Multiplier

The multiplier can multiply a no's having bit size = 2

Product bit size will be sum of bitsize of i/p

$$\text{i.e. } 2 + 2 = 4$$

Max range of its o/p is $3 \times 3 = 9$. So we can accomodate decimal 9 in 4 bits.

eg $\rightarrow$

$A_1 A_0$
 $B_1 B_0$

$\boxed{B_0 A_1} \quad \boxed{A_0 B_0} \leftarrow \text{Partial Product}$

$\boxed{B_1 A_1} \quad \boxed{B_1 A_0}$

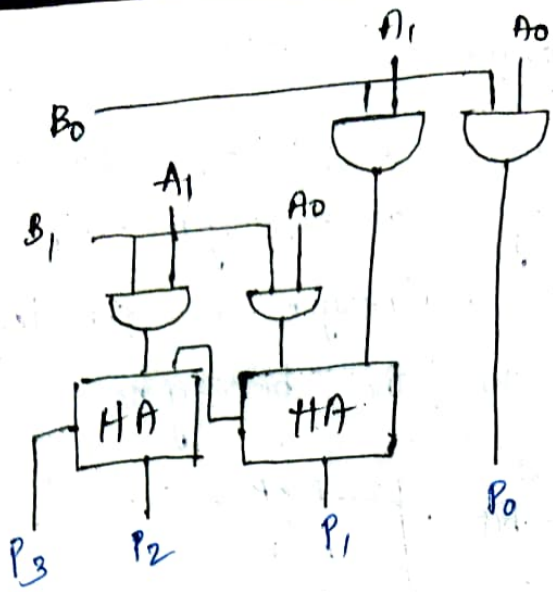
$P_3 \quad P_2 \quad P_1 \quad P_0$

$$P_0 \rightarrow A_0 B_0$$

$$P_1 \rightarrow B_1 A_0 + B_0 A_1$$

$$P_2 \rightarrow B_1 A_1 + \text{carry of } P_1$$

$$P_3 \rightarrow \text{carry of } P_2$$



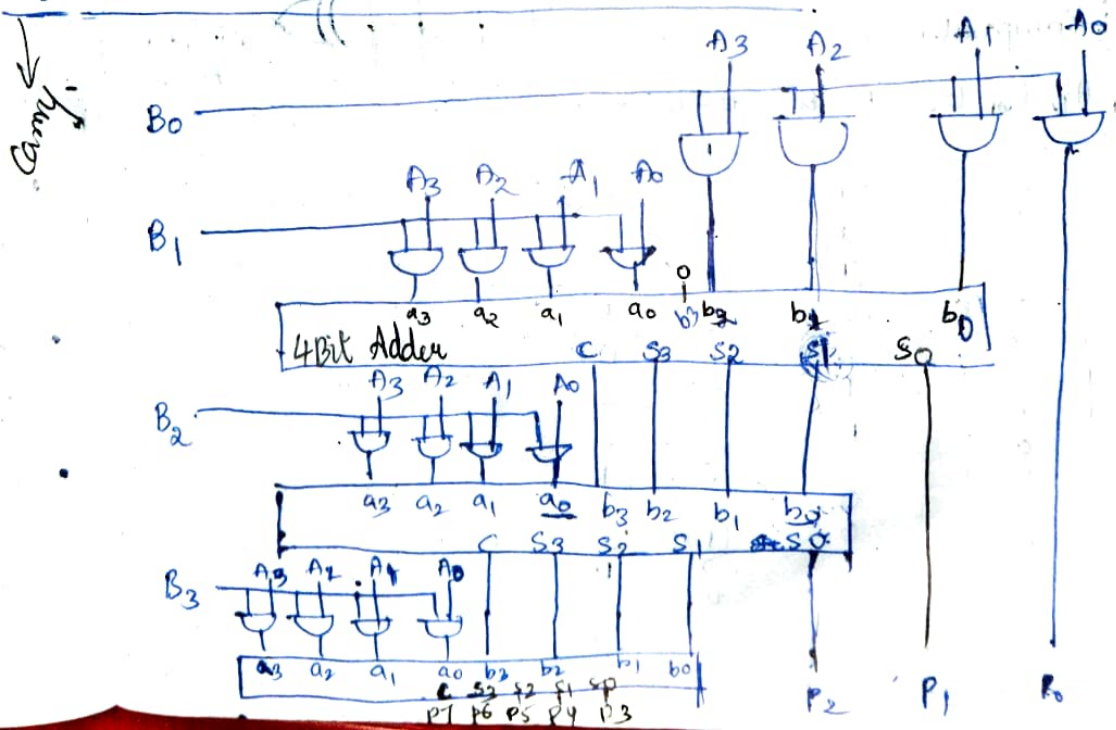
Multiplication by
and gates
addition carried by
half adders.

4x4 Multiplier

$A_3 \quad A_2 \quad A_1 \quad A_0$
 $B_3 \quad B_2 \quad B_1 \quad B_0$

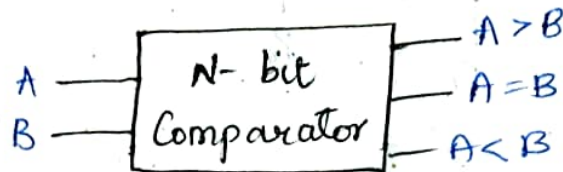
$a_3 \quad a_2 \quad a_1 \quad a_0$
 $b_3 \quad b_2 \quad b_1 \quad b_0$
 $S_3 \quad S_2 \quad S_1 \quad S_0$
 $C_3 \quad C_2 \quad C_1 \quad C_0$

$B_0 A_3 \quad B_0 A_2 \quad B_0 A_1 \quad B_0 A_0$
 $B_1 A_3 \quad B_1 A_2 \quad B_1 A_1 \quad B_1 A_0$
 $B_2 A_3 \quad B_2 A_2 \quad B_2 A_1 \quad B_2 A_0$
 $B_3 A_3 \quad B_3 A_2 \quad B_3 A_1 \quad B_3 A_0$



2 Bit Magnitude Comparator

Magnitude Comparator :- is a combinational circuit that compares two digital or binary no's in order to find out whether 1 binary no is equal less than or greater than the other binary no.



1 Bit Comparator

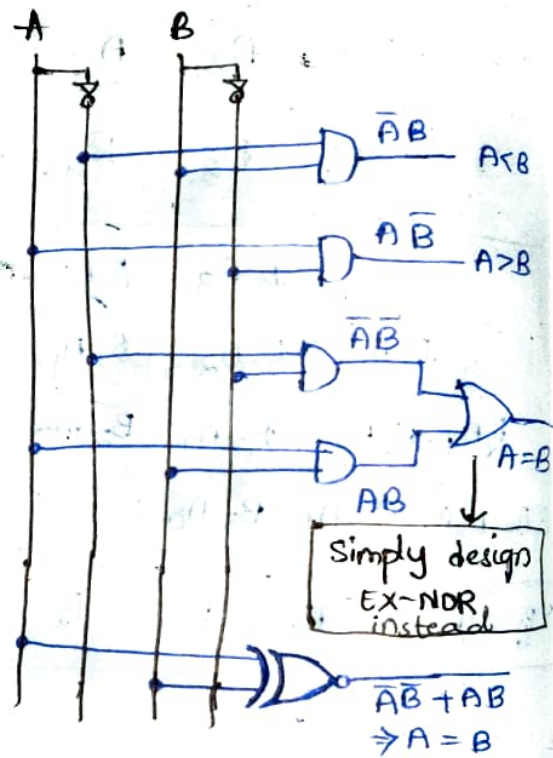
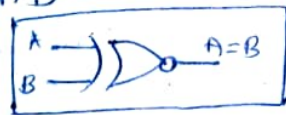
A	B	A < B	A > B	A = B
0	0	0	0	1
0	1	1	0	0
1	0	0	1	0
1	1	0	0	1

$$A < B \Rightarrow \bar{A} \cdot B$$

$$A > B \Rightarrow A \cdot \bar{B}$$

$$A = B \Rightarrow \bar{A}\bar{B} + AB$$

$$= A \odot B$$



2-Bit Comparator

A ₁	A ₀	B ₁	B ₀	A < B	A = B	A > B
----------------	----------------	----------------	----------------	-------	-------	-------

0	0	0	0	0	1	0
0	0	0	1	0	0	1
0	0	1	0	1	0	0
0	0	1	1	0	1	0
0	1	0	0	1	0	0
0	1	0	1	0	0	1
0	1	1	0	0	1	0
0	1	1	1	0	1	0
1	0	0	0	0	0	1
1	0	0	1	0	0	1
1	0	1	0	0	0	1
1	0	1	1	0	0	1
1	1	0	0	0	0	1
1	1	0	1	0	0	1
1	1	1	0	0	0	1
1	1	1	1	0	1	0

$A_1 A_0$	$B_1 B_0$	00	01	11	10
00	0	1	1	1	1
01	0	0	1	1	1
11	0	0	0	0	0
10	0	0	1	0	0

$A < B$

$$\bar{A}_1 B_1 + \bar{A}_1 \bar{A}_0 B_0 + \bar{A}_0 B_1 B_0$$

$A_1 A_0$	$B_1 B_0$	00	01	11	10
00	1	0	0	0	0
01	0	1	0	0	0
11	0	0	1	0	0
10	0	0	0	1	1

$$\bar{A}_1 \bar{A}_0 B_1 \bar{B}_0 + \bar{A}_1 \bar{A}_0 \bar{B}_1 B_0 +$$

$$A = B \Rightarrow A_1 A_0 B_1 B_0 + A_1 \bar{A}_0 B_1 \bar{B}_0 +$$

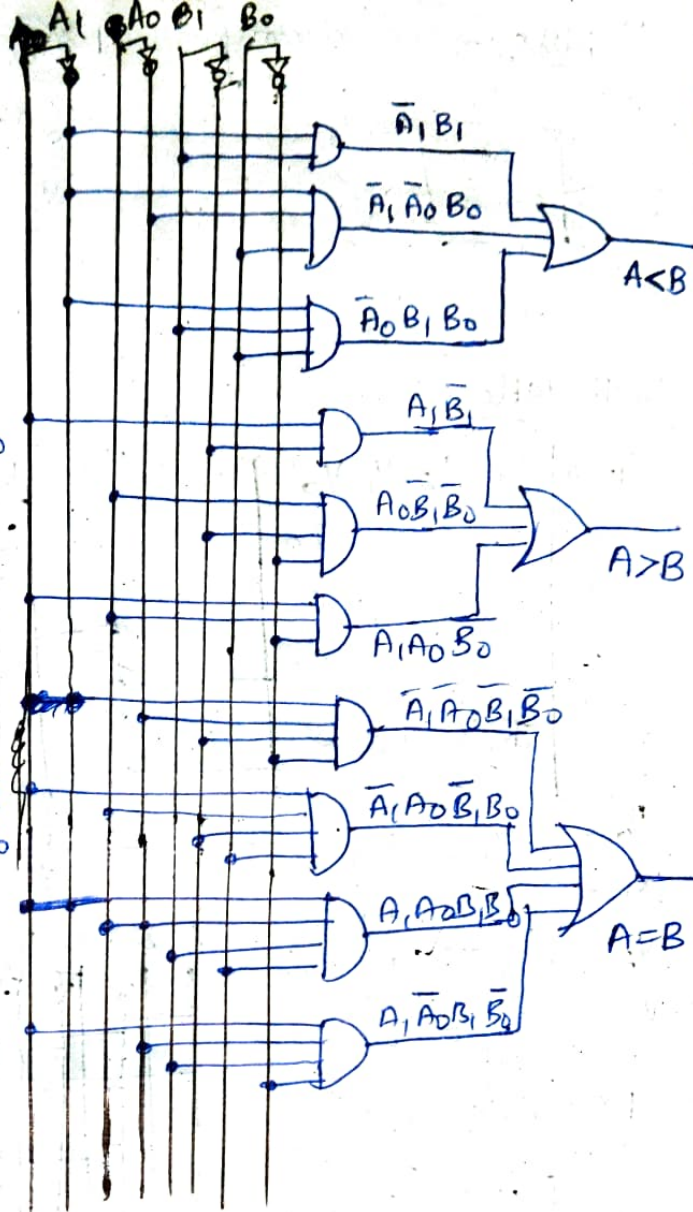
$A_1 A_0$	$B_1 B_0$	00	01	11	10
00	0	0	0	0	0
01	1	0	0	0	0
11	1	1	1	0	1
10	1	1	1	0	0

$$A_1 \bar{B}_1 \bar{B}_0 + A_0 \bar{B}_1 \bar{B}_0 + A_1 A_0 \bar{B}_0$$

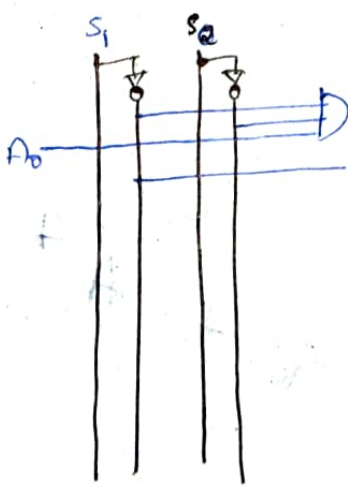
4x1 Mux



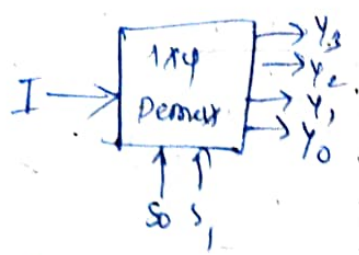
S_1	S_0	Y
0	0	A_0
0	1	A_1
1	0	A_2
1	1	A_3



$$Y = \bar{S}_1 \bar{S}_0 A_0 + \bar{S}_1 S_0 A_1 + S_1 \bar{S}_0 A_2 + S_1 S_0 A_3$$



Demux :- one to many

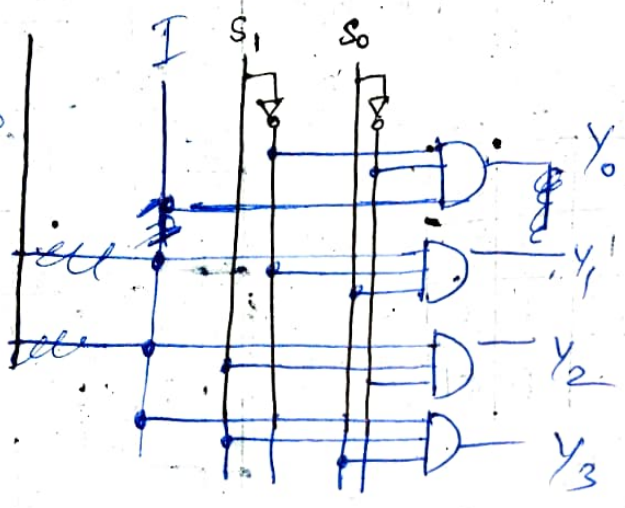


1x4 demux
 2^n - select lines
 2^2 - s_0, s_1

1x8 demux
 2^n - select lines
 2^3 - s_0, s_1, s_2

Truth Table

s_1	s_0	y_3	y_2	y_1	y_0
0	0	0	0	0	I_0
0	1	0	0	I_1	0
1	0	0	I_2	0	0
1	1	0	0	0	0



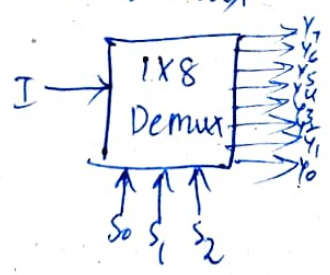
$$y_0 = \bar{s}_1 \bar{s}_0 I_0$$

$$y_1 = \bar{s}_1 s_0 I_1$$

$$y_2 = s_1 \bar{s}_0 I_2$$

$$y_3 = s_1 s_0 I_3$$

1x8 Demux



s_0	s_1	s_2	y_7	y_6	y_5	y_4	y_3	y_2	y_1	y_0
0	0	0	0	0	0	0	0	0	0	I_0
0	0	1	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	0
0	1	1	0	0	0	0	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0
1	0	1	0	0	0	0	0	0	0	0
1	1	0	0	0	0	0	0	0	0	0
1	1	1	0	0	0	0	0	0	0	0

2) A combinational circuit is specified by the following Boolean Function

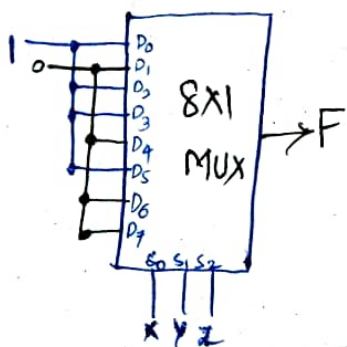
$$F(A, B, C) = \sum m(2, 5, 6), \quad f_2(A, B, C) = \sum m(1, 3, 4)$$

Implement the circuit with a 3x8 decoder and OR gates. Use block diagram for the decoder.

Implement the following functions using 8:1 MUX

$$F(x, y, z) = \sum m(0, 2, 3, 5)$$

x	y	z	F
s_2	s_1	s_0	F
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	0



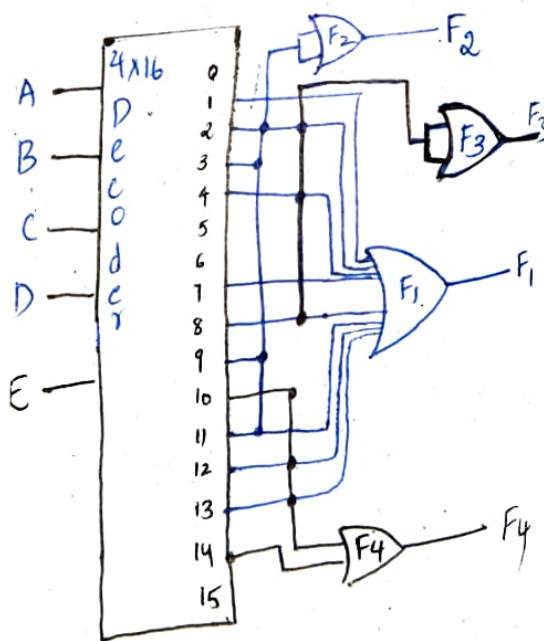
Implement the following multiple o/p combinational logic circuit using a 4 line to 16-line decoder.

$$F_1 = \sum m(1, 2, 4, 7, 8, 11, 12, 13)$$

$$F_2 = \sum m(2, 3, 9, 11)$$

$$F_3 = \sum m(10, 12, 13, 14)$$

$$F_4 = \sum m(2, 4, 8)$$



If $E = 0$ then replace D with D_0

Implement the func. $F(A, B, C, D) = \sum m(2, 3, 4, 5, 8, 9, 12)$ using 16X1 mux and 8X1 mux.

	D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7
A'	0	1	2	3	4	5	6	7
A	8	9	10	11	12	13	14	15
	A	A'	A'	A'	1	A'	0	0

